

Callum O'Neill

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Outline

Highly motivated Mechanical Engineering MEng student with hands-on experience in design, CAD, and embedded systems. Currently achieving a first-class standard at the University of Nottingham, with proven leadership and workplace skills gained through both academic projects and professional experience. Strong expertise in mechanical design, statics, dynamics, material choice, thermodynamics, fluid dynamics, electronics, and CAD (SolidWorks CSWA certified). Skilled at delivering multidisciplinary engineering solutions and collaborating effectively within teams.

Education

University of Nottingham

September 2023 - Present

Mechanical Engineering MEng | Achieving a first class

Key Modules:

Statics and Mechanics | Dynamics of Systems | Thermodynamics and Fluid Dynamics | Mathematics | Materials in Design | Computer Engineering and Mechatronics | Computer Modelling Techniques (CFD) | Computer Aided Engineering (CAE) | Biomechanics.

CanSat Project

September 2025 – Present

Electronics – Team lead

- Leading the electronics sub-team in the design and integration of embedded systems for a CanSat satellite constrained to 115 mm diameter by 95 mm height (as per the constraints of the given rocket).
- Developing C-based embedded software for atmospheric pressure and temperature data acquisition, sensor integration, and onboard image capture.
- Supporting system testing, fault-finding, and validation to ensure mission requirements of ~2 km deployment altitude and 10 m ground traversal are met.
- Collaborating across mechanical, electronics, and software teams to deliver a fully integrated system.

University Design Project

January 2024 - May 2024

Pneumatic tri-piston motor - Team lead

- Led a multidisciplinary team in the design, analysis, and manufacture of a tri-piston pneumatic air motor.
- Performed torque, efficiency, and performance calculations to validate operation at 4 bar air pressure. Applied mechanical design principles, tolerancing, and fits to ensure manufacturability and reliable assembly.

- Produced detailed SolidWorks CAD models and engineering drawings.
- Manufactured components using workshop machinery (lathe, milling, drilling) and 3D printing, followed by functional testing.

Lawrence Sheriff School

September 2017 - June 2023

A-Levels: Mathematics (A), Physics (B), Engineering (BTEC: Distinction*), Personal Finance (AS-Level: B)

GCSEs: 10 GCSEs (8-6), including Maths (8) and English Literature/Language (7, 8) and a Level 3 EPQ (A) on the sustainability of electric cars.

Experience

Anker Elemental

September 2025 - Present

Water treatment specialist

- Gained hands-on experience with pumps and fluid circulation systems and heat exchange equipment, developing a practical understanding of real-world thermofluidic applications.
- Performed maintenance on industrial water systems, including evaporative cooling towers and storage tanks, ensuring compliance with safety and environmental standards and performance optimisation.
- Conducted chemical analysis and diagnostics to identify faults, improve system reliability, and reduce operational downtime.
- Developed problem-solving skills through real-world engineering maintenance.

Curry's

September 2022 - August 2023

Sales assistant/Warehouse operative

- Delivered technical product advice to customers, strengthening communication and customer-focused problem solving.
- Worked effectively in a fast-paced team environment to meet performance targets.
- Managed stock intake, inventory control, and order picking under strict time constraints.

Technical skills

- CAD & Design: SolidWorks (CSWA Certified), engineering drawings, tolerancing (GD&T).
- Programming: MATLAB, C, Arduino, embedded systems.
- Engineering Analysis: Thermodynamics, fluid dynamics, CFD, CAE, statics, dynamics.
- Manufacturing: Lathe, milling, drilling, 3D printing. Testing & Validation: System testing, fault-finding, performance analysis

References available on request.